

INTERNAL ASSESSMENT

MATHEMATICS ANALYSIS AND APPROACHES HL

TOPIC: Optimum Tilt Angle for Photovoltaic Solar Panels

PAGE COUNT: 18

Aim

The aim is to determine the optimal tilt angle for solar panels required for efficient solar energy absorption. Through principles of trigonometry, calculus, and geometry, this study aims to explore the relationship between solar panel orientation, incident sunlight angle, and energy yield.

Introduction

As the demand for renewable energy grows, solar power stands out as a promising solution. Solar panels, also known as photovoltaic panels, harness the power of sunlight to generate electricity through the photovoltaic effect¹. This process involves the conversion of sunlight directly into electricity using semiconducting materials², typically silicon. Solar energy, as an alternative to traditional fossil fuels, offers numerous benefits including renewable and abundant availability, reduced greenhouse gas emissions³, and potential cost savings over time. The efficiency of solar panels relies on factors such as tilt angle and exposure to sunlight⁴. In this investigation, I undertake a mathematical analysis to determine the optimal tilt angle for solar panels, aiming to maximize energy absorption. While both households and complexes are increasingly adopting solar energy methods, their orientation and positioning can vary widely. Large solar power plants often boast thousands of panels capable of adjusting orientation to track the sun⁵, optimizing energy absorption. However, many households rely on fixed installations, necessitating careful consideration of tilt angle and surface area to optimize energy output.

This investigation will assume the solar panel will only be tilted in one direction, there is nothing present between the solar panel and the sun, and one day is 24 hours. Initially, we will assume the earth has no axial tilt, whereas in reality it has an axial tilt of 23.5°. The process to calculate the optimum tilt angle will remain the same.

¹ Green, Martin A. *Solar Cells: Operating Principles, Technology, and System Applications*. Prentice-Hall, 1982.

² Nelson, Jenny. *The Physics of Solar Cells*. Imperial College Press, 2003.

³ IPCC. *Climate Change 2021: The Physical Science Basis*. Cambridge University Press, 2021.

⁴ Duffie, John A., and William A. Beckman. *Solar Engineering of Thermal Processes*. 4th ed., Wiley, 2013.

⁵ Mousazadeh, Hossein, et al. "A Review of Principle and Sun-Tracking Methods for Maximizing Solar Systems Output." *Renewable and Sustainable Energy Reviews*, vol. 13, no. 8, 2009, pp. 1800–1818.

Global Significance and Rationale.

Finding the optimum angle helps enhance the design and implementation of more efficient and cost-effective solar energy systems, thereby advancing the transition towards a cleaner and more sustainable energy future.

I chose to investigate this topic due to its relevance in modern day society. With the increase in global warming, climate change has become one of the biggest issues in the world. Finding suitable alternatives to fossil fuels is of utmost importance, and solar energy serves the purpose of a replaceable, renewable source of energy. After witnessing the installation of solar panels in my grandparents' house, I was intrigued as to how they decided where to put the photovoltaic panels. Hence, I decided to investigate the optimum tilt angle for solar panels, and eventually aim to find the optimum tilt angle for their house terrace.

Mathematical context and overview

First, we will assume the earth as a spherical object and no axial tilt. angle between a planet's rotational axis and the plane of its orbit, also known as the orbital plane. The Earth is actually tilted 23.5° .⁶ It can be represented on a 3D xyz coordinate system, as shown in figure 1.

The sun is on the negative x-axis and the centre of the earth intersects the x,y,z axis. Since the Axial tilt is assumed to be 0° , the rotational axis for earth is along the y-axis.

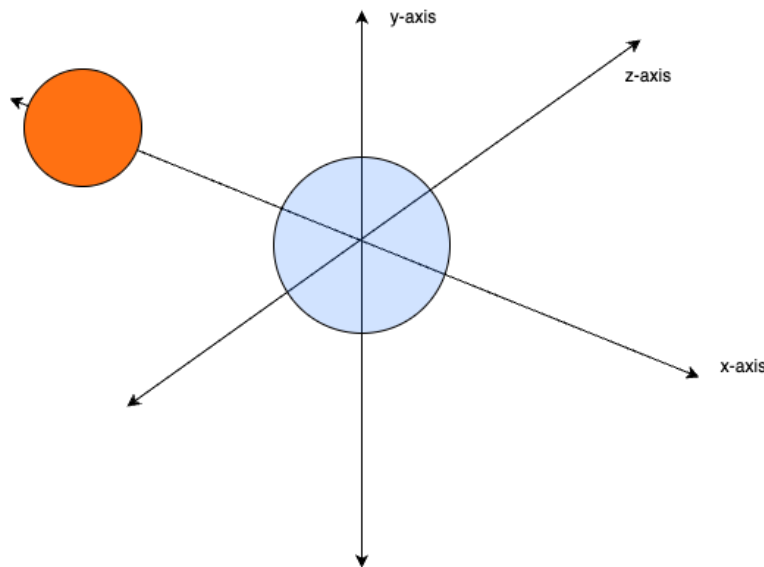


Figure 1: Earth in 3D x,y,z coordinate system.

Figure 2 illustrates a 2D image of the earth along with the x, z axis. The earth is viewed along the rotation of axis, or the y-axis. Time, in hours, can be represented as angular displacement from point A. At midnight, the point will be furthest away from the sun, hence represented by point A. This is taken as an angular displacement of 0 radians. As the day passes, the earth will rotate clockwise. After 12 hours from midnight, half the day is past and the sun has rotated to an angular displacement of π radians.

⁶ Axial Tilt - Energy Education. energyeducation.ca/encyclopedia/Axial_tilt.

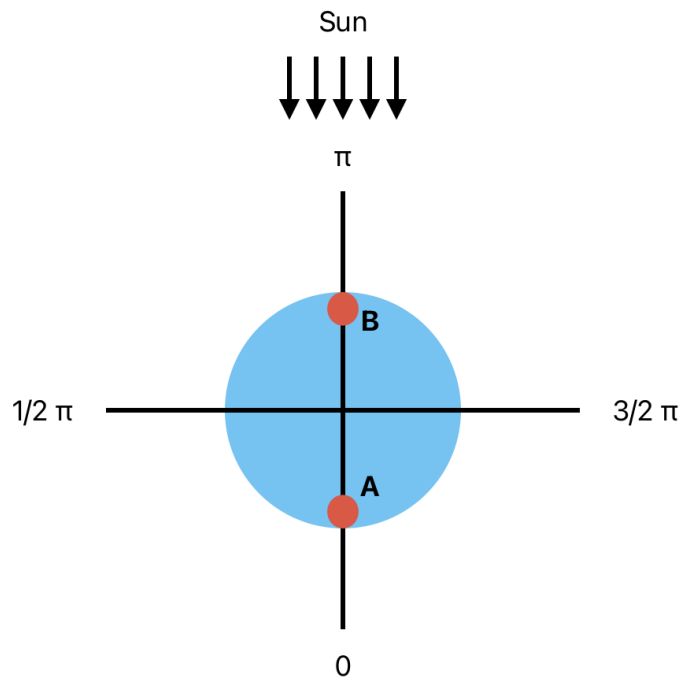


Figure 2: 2D image of earth with sun rays

An axial tilt gives rise to seasons, which vary the amount of sunlight received per day during a full year. This would require further modelling, and is beyond the scope of this investigation.

As the axial tilt was assumed as 0 for this investigation, all days throughout the year will receive the same amount of sunlight per day. Hence, to simplify the problem, we can calculate the optimum tilt angle for maximum energy absorption in one day, which is one full rotation.

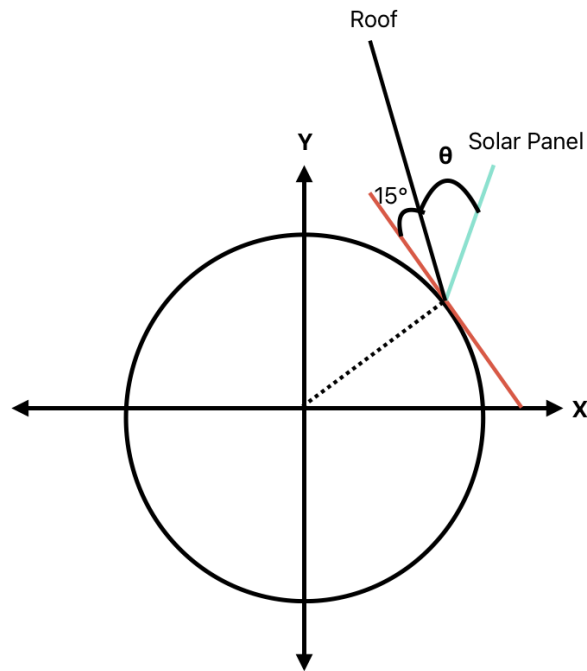


Figure 3: Representation of tilt angle on Earth.

Figure 3 illustrates the angle θ considered for the optimization. For any point on earth, a tangent is drawn. Placing a solar panel flat on the ground is equivalent to placing it along the tangent drawn. Hence, the angle considered is the angle between the tangent and the solar panel. My grandparents' house has a roof incline of approximately 15° , which will be accounted for when finding the optimum tilt angle for their terrace. This angle was given by the maintenance team at their locality and included in the description of their villa.

Methodology

This investigation first aims to find the optimum fixed tilt angle by modelling the situation using vectors and trigonometry. Through simplified diagrams and appropriate assumptions to fit the situation of my grandparents' house, a function will be derived to represent the amount of sunlight absorbed by the solar panel using the limit definition of the integral. Next, differentiation (optimization) will be employed to calculate the stationary points (maxima) to find the optimum point.

MODELLING

Section I: Finding the sunlight received by the Solar Panel

The amount of solar energy generated by the solar panel will depend on the amount of sunlight received by the solar panel. Since the rays of the sun are scattered and change directions throughout the day while the position of the panel, in this case, remains constant, there will be a many orientations of rays. To combat this, the rays of the sun can be visualized using vectors. The amount of sunlight received by the panel is the perpendicular component of the vector. It is assumed that all vectors are of same magnitude and direction and the magnitude of the sunlight ray vector is 1, which are reasonable assumptions. This can be illustrated in Figure 4. This diagram, as well as following diagrams, were drawn using FreeForm.⁷

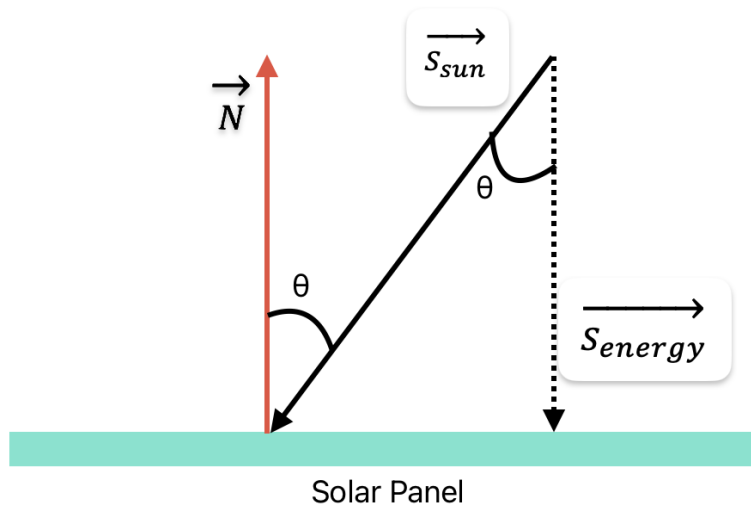


Figure 4: Perpendicular Vector of sunlight

Vector N is the normal vector to the solar panel. To find the amount of sunlight received by the solar panel, we must find the magnitude of the perpendicular vector, S_{energy} .

$$\cos \theta_{ray} = \frac{\left| \overrightarrow{S_{energy}} \right|}{\left| \overrightarrow{S_{sun}} \right|} \Leftrightarrow \left| \overrightarrow{S_{energy}} \right| = \cos \theta_{ray} \times \left| \overrightarrow{S_{sun}} \right|$$

⁷ App Store. "Freeform." App Store, 25 Oct. 2022, apps.apple.com/us/app/freeform/id6443742539.

where the $|x|$ represent the magnitude of any given vector x .

Since $\left| \overrightarrow{s_{sun}} \right| = 1$ was a previous assumption,

$$\cos \theta_{ray} = \left| \overrightarrow{s_{energy}} \right|$$

Since intuitively the angle θ can only reside between 0 and $\frac{\pi}{2}$, the magnitude of S_{energy} will always be greater than 0.

To calculate the value of $\cos \theta$, we will find the dot product of Vector N and Vector S

$$if \ \vec{N} = \begin{Bmatrix} N_x \\ N_y \\ N_z \end{Bmatrix} \text{ and } \vec{s_{sun}} = \begin{Bmatrix} S_x \\ S_y \\ S_z \end{Bmatrix}$$

$$\vec{N} \cdot \vec{s} = N_y S_y + N_x S_x + N_z S_z = \left| \vec{N} \right| \left| \vec{s} \right| \cos \theta_{ray}$$

$$\cos \theta_{ray} = \frac{N_y S_y + N_x S_x + N_z S_z}{\left| \vec{N} \right| \left| \vec{s_{sun}} \right|}$$

Since the magnitude of $\vec{s_{sun}} = 1$ and the sun lies on the negative x-axis as shown in figure one, $\vec{s_{sun}}$ can be represented as:

$$\vec{s_{sun}} = \begin{Bmatrix} -1 \\ 0 \\ 0 \end{Bmatrix}$$

Now that we have $\vec{s_{sun}}$, we need to find the vector $\vec{N}$ in order to calculate the value of $\cos \theta$:

Since earth is a 3D object, the vector must contain 3 components – x,y,z coordinates. First, we will generalize the representation for Vector V, which forms the angle of latitude with the Vector V_2 . Vector V lies on the plane of Vector V_2 and the y-axis, and Vector V_2 lies on the xz-axis with the angle between their components representing the angular displacement (time).

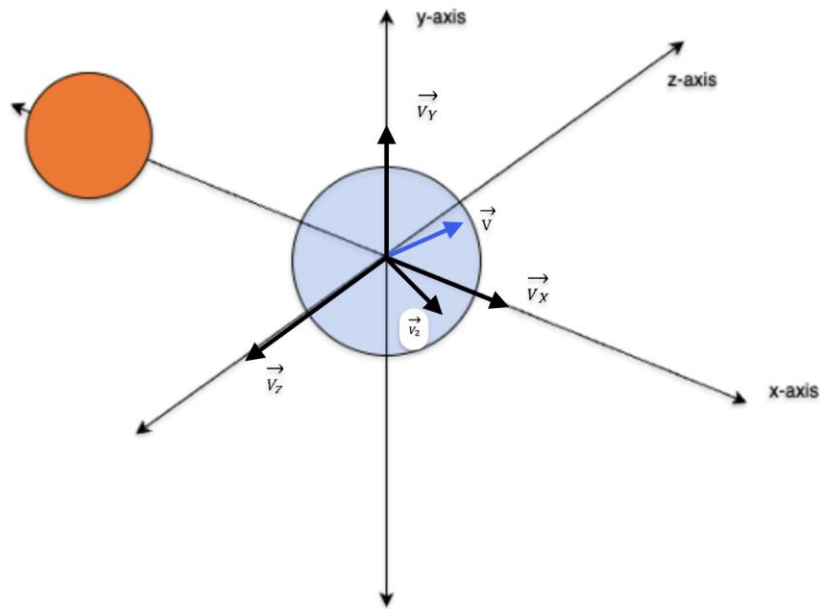


Figure 5: Components of Vector V

Note: Magnitudes of components are not drawn to scale.

This can be determined by viewing the earth from different planes, as illustrated in Figure 6 and 7. In Figure 6, the view of the plane containing the y-axis seen in Figure 1 and $\vec{V}$. It also showcases the radius of the earth. The magnitude of $\vec{V}$ is taken as the radius of the earth. The angle between the Vector V_2 and $\vec{V}$ is the latitude. My grandparents' house has a latitude of 28.4595° . In Figure 7, the xz plane is viewed.

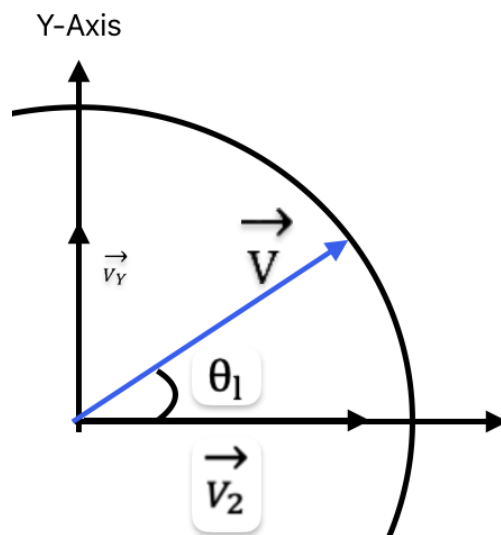


Figure 6: Component Y

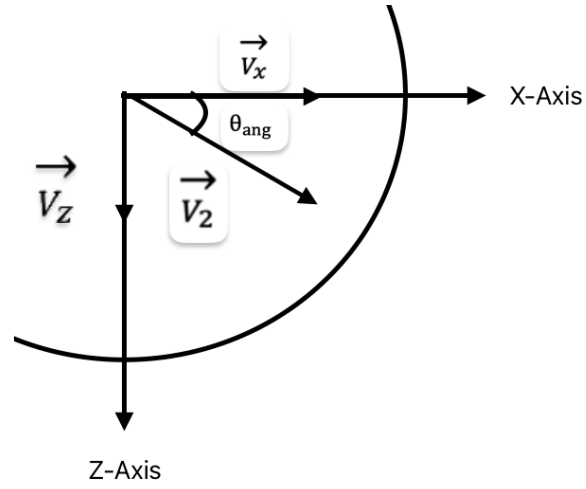


Figure 7: Component X,Z

Using information in Figure 6 and 7, the following can be derived:

$$\text{Radius of the Earth} = r = |\vec{r}|$$

$$\vec{r}_{V_2} = r \cos \theta_l$$

$$\vec{r}_{V_Y} = r \sin(\theta_l)$$

$$\vec{r}_{V_x} = \vec{r}_{V_2} \cos \theta_{ang} = r \cos(\theta_l) \cos \theta_{ang}$$

$$\vec{r}_{V_z} = \vec{r}_{V_2} \sin \theta_{ang} = r \cos(\theta_l) \sin \theta_{ang}$$

Hence, the vector $\vec{r}$ can be represented as $\begin{Bmatrix} r \cos(\theta_l) \cos \theta_{ang} \\ r \sin(\theta_l) \\ r \cos(\theta_l) \sin \theta_{ang} \end{Bmatrix}$

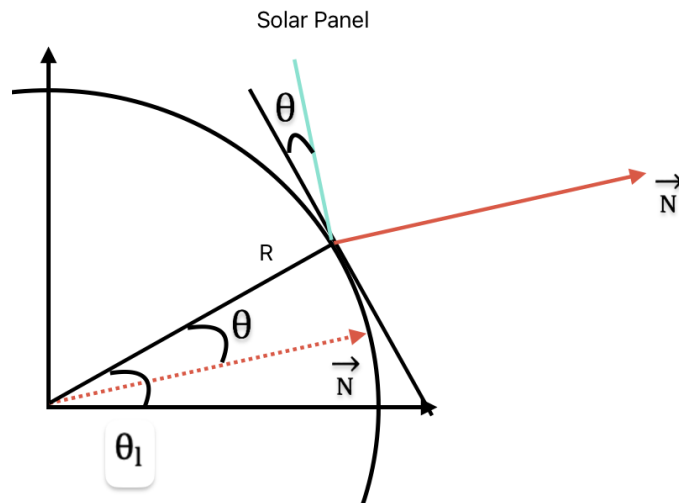


Figure 8

Using the generalization found above, we can find a vector representing the normal vector by substituting the angle of latitude for $\theta_l - \theta$ as explained in Figure 8. The angle between $\vec{N}$ and r (which can also be represented as Vector V) is the same as the tilt angle, found using the properties of angles. Subtracting this from the latitude gives us the angle formed if the normal vector had a magnitude the size of the radius, and translated to the origin. Hence, the same generalization found earlier may be used to find the vector representation of the normal.

Hence, the vector $\vec{N}$ can be represented as
$$\begin{pmatrix} r \cos(\theta_l - \theta) \cos \theta_{ang} \\ r \sin(\theta_l - \theta) \\ r \cos(\theta_l - \theta) \sin \theta_{ang} \end{pmatrix}$$

Recalling the formula we derived to find the amount of sunlight received by the solar panel

$$\cos \theta_{ray} = \frac{N_y S_y + N_x S_x + N_z S_z}{|\vec{N}| |\vec{S_{sun}}|}$$

Therefore, We can plug in the values of $|\vec{N}|$ and $|\vec{S_{sun}}|$:

$$\cos \theta_{ray} = \frac{-(N_y S_y + N_x S_x + N_z S_z)}{|\vec{N}|} = \frac{-(r \cos(\theta_l - \theta) \cos \theta_{ang})}{r} = -\cos(\theta_l - \theta) \cos \theta_{ang}$$

The formed equation in terms of angular displacement, θ_{ang} , can be expressed as:

$$f(\theta_{ang}) = -\cos(\theta_l - \theta) \cos \theta_{ang}$$

This equation allows us to determine the amount of sunlight received by PV panels for a given tilt angle, latitude and time (angular displacement). However, we require to find the total amount of sunlight in a day to find the optimal tilt angle. This is done in Section II.

Section II: Calculating the total sunlight received by the Solar Panel in a day.

The total amount of sunlight received during the day can be found taking short time intervals, and multiplying this by amount of sunlight.

$$Total\ Sunlight = f(\theta_{ang}) \times \Delta\theta_{ang}$$

where θ_{ang} corresponds to a small increment in angular displacement.

Since the axial tilt in this case is 0, the day will have a standard sunrise time at an angular displacement of $\frac{\pi}{2}$ radians and a standard sunset time at an angular displacement of $\frac{3\pi}{2}$ radians, between which the solar panels will be exposed to sunlight. Adding up the values of total sunlight from $\theta_{ang} = \frac{\pi}{2}$ radians to $\theta_{ang} = \frac{3\pi}{2}$ radians:

$$Total\ Sunlight = \sum_{i=0}^n f\left(\frac{\pi}{2} + (\Delta\theta_{ang} \times i)\right) \times \Delta\theta_{ang}$$

where n = number of time intervals between $\theta_{ang} = \frac{\pi}{2}$ radians to $\theta_{ang} = \frac{3\pi}{2}$ radians, and $\Delta\theta_{ang}$ = the time of each time interval.

From this we can derive that:

$$\Delta\theta_{ang} = \frac{\frac{3\pi}{2} - \frac{\pi}{2}}{n}$$

Substituting this value of $\Delta\theta_{ang}$ into the initial equation:

$$Total\ Sunlight \approx \sum_{i=0}^n f\left(\frac{\pi}{2} + \left(\frac{\frac{3\pi}{2} - \frac{\pi}{2}}{n} \times i\right)\right) \times \frac{\frac{3\pi}{2} - \frac{\pi}{2}}{n}$$

Since the exact amount of total sunlight will need an infinite number of time intervals (n), taking the limit of this function to n approaches infinity will help us achieve the exact function.

$$Total\ Sunlight = \lim_{n \rightarrow \infty} \sum_{i=0}^n f\left(\frac{\pi}{2} + \left(\frac{\frac{3\pi}{2} - \frac{\pi}{2}}{n} \times i\right)\right) \times \frac{\frac{3\pi}{2} - \frac{\pi}{2}}{n}$$

Using the limit definition of integral, explained below, helps us represent this function in terms of an integral, rather than an infinite sum, which is more convenient when differentiating to find the optimum point (maxima).

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \frac{b-a}{n}$$

where x_k^* is the midpoint of the k^{th} interval of $[a, b]$ having length $(b-a)/n$, namely

Figure 9: Explaining limit definition of integral⁸

Comparing the 2 equations to find a function in terms of θ ,

$$Total\ Sunlight = f(\theta) = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} f(\theta_{ang}) d\theta_{ang} = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} -\cos(\theta_l - \theta) \cos \theta_{ang} d\theta_{ang}$$

Since $-\cos(\theta_l - \theta)$ is a constant, we can simplify this further:

$$\begin{aligned} f(\theta) &= -\cos(\theta_l - \theta) \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos \theta_{ang} d\theta_{ang} \\ &= -\cos(\theta_l - \theta) [\sin \theta_{ang}]_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \\ &= -\cos(\theta_l - \theta) \left(\sin \frac{3\pi}{2} - \sin \frac{\pi}{2}\right) \end{aligned}$$

⁸ The Limit Definition of a Definite Integral. www.math.ucdavis.edu/~kouba/CalcTwoDIRECTORY/defintdirectory/DefInt.html.

$$= -\cos(\theta_l - \theta) (-1 - 1)$$

Therefore, the equation for total amount of sunlight received in one day is:

$$\text{Total amount of sunlight} = 2 \cos(\theta_l - \theta)$$

OPTIMIZING THE FUNCTION

To find the point at which the amount of sunlight, and hence energy produced, is greatest, we can find the maximum point. The maxima is a turning point (stationary point) where the gradient of the tangent is equal to 0.

To do this, we can differentiate the function, equate to 0 and hence find the optimum angle, θ , or we can plot the function and identify the turning point.⁹

Method 1: Differentiation Method

Let:

$$y = 2 \cos(\theta_l - x)$$

where x = tilt angle of solar panels, θ .

$$\frac{dy}{dx} = -2 \sin(\theta_l - x) \times -1 = 2 \sin(\theta_l - x)$$

Equating $\frac{dy}{dx} = 0$ to find stationary point:

$$2 \sin(\theta_l - x) = 0$$

$$\sin(\theta_l - x) = 0$$

$$(\theta_l - x) = \sin^{-1} 0$$

$$(\theta_l - x) = 0, \pi, 2\pi, 3\pi, \dots$$

$$(\theta_l - x) = k\pi \quad k \in \mathbb{Z}$$

$$x = \theta_l - k\pi \quad k \in \mathbb{Z}$$

Since $x \in [0, \frac{\pi}{2}]$,

$$x = \theta_l$$

⁹ Musolino, Larry. "4.4 Optimization for Functions." Pressbooks, 1 Feb. 2021, psu.pb.unizin.org/math110/chapter/4-4-optimization-for-functions.

Hence, the optimum tilt angle for solar panels is the angle of the latitude. Recalling that the latitude of my grandparents' house is 28.4595° , and the angle of inclination of the roof is approximately 15° . Hence, when installing the solar panels the optimum angle of installation is $28.4595^\circ - 15^\circ \approx 13.5^\circ$

Method 2: Graph Method

Plotting the equation in desmos,

$$y = 2 \cos(0.489 - x)$$

Where 0.489 is the latitude of Gurgaon, India converted to radians, x = tilt angle, y = total amount of sunshine.

The graph drawn below has total amount of sunshine on the y-axis and tilt angle of solar panel on the x-axis.

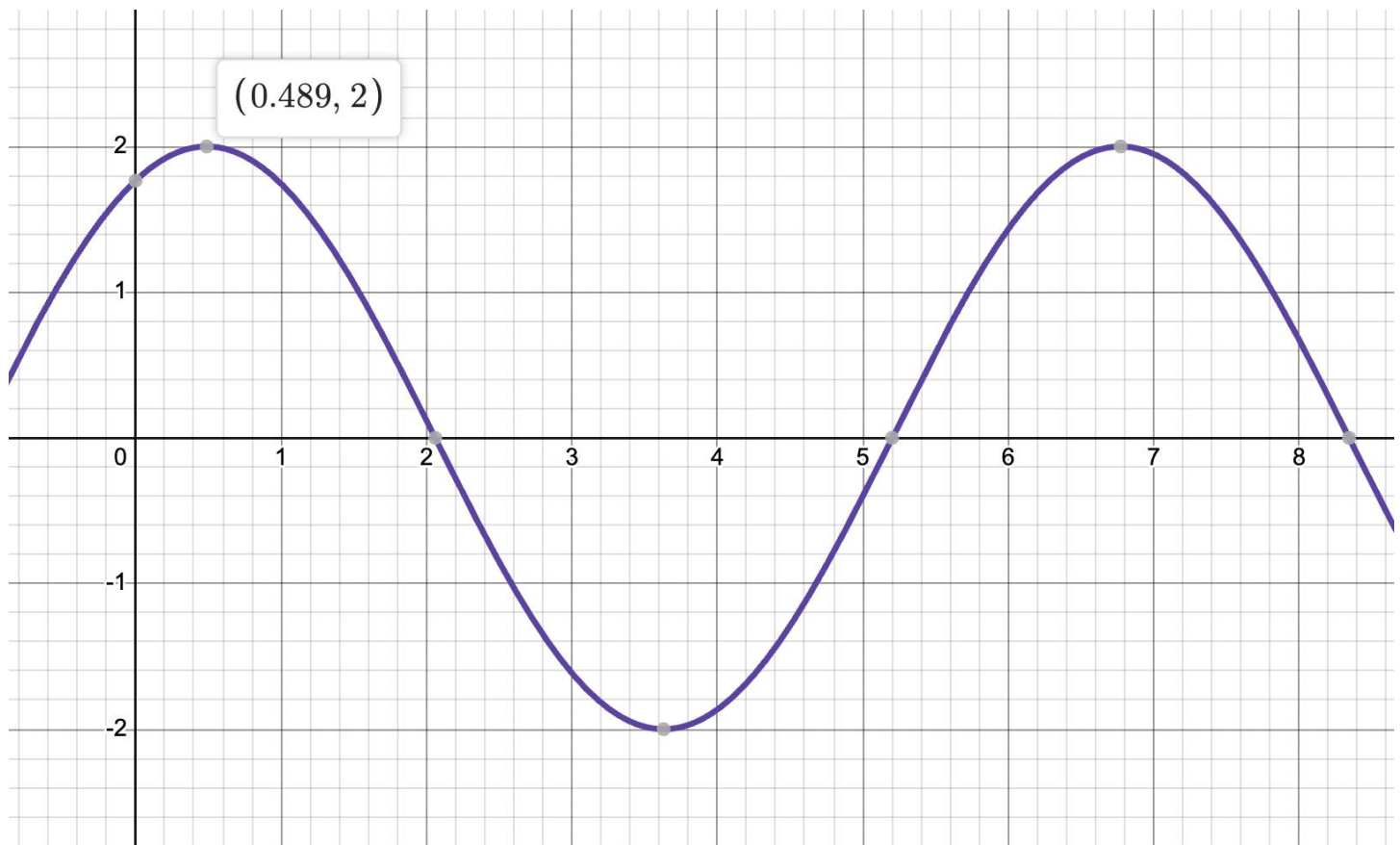


Figure 10: Graph of amount of sunshine v/s tilt angle

As seen in the graph, the maximum point is at the tilt angle, represented by the x-value of 0.489, is the same as the latitude of the city. (Converting 0.489 radians gives us 28°)

Conclusion and Evaluation

After conducting the mathematical exploration to determine the optimum tilt angle for solar panels at my grandparent's house. I concluded that the optimum angle is 0.489 radians, or 28 degrees, which is equal to the latitude of the city. However, taking roof tilt into account, this would decrease by 15 degrees in the case of my grandparents' house, as seen earlier. This finding aligns with the theoretical principle that, assuming an axial tilt of 0, the optimal tilt angle for maximizing solar efficiency is equal to the location's latitude. The investigation successfully combined various factors such as time, latitude, and total sunlight into a single equation, which was then differentiated to find the optimal angle. The use of trigonometry and vectors to model the situation was a significant strength, as it simplified the problem and provided a clear and well-explained solution. Furthermore, the study was tailored specifically to my grandparents' house, adding a practical and personalized dimension to the exploration. To ensure the robustness of the results, the optimization process was carried out using two different methods, both of which supported the conclusion.

However, the study is not without its limitations. One major limitation is the assumption that the axial tilt is 0, which implies that the amount of sunlight received is constant throughout the year. In reality, the Earth's axial tilt is approximately 23.5 degrees, leading to seasonal variations in sunlight. This simplification affects the accuracy of the results, and incorporating the axial tilt could be a valuable extension of the study. Additionally, the accuracy of the results is limited by the assumptions made regarding the measurement of the rooftop's tilt angle. Any inaccuracies in these measurements could lead to deviations from the most optimal tilt angle. Another limitation is the assumption that the Earth's orbit around the sun is circular, which allows for the use of angular measure as a representation of time. However, the orbit is actually elliptical, introducing further inaccuracies into the results. Despite these limitations, the study provides a solid foundation for understanding the relationship between latitude and solar panel efficiency, and it highlights areas for further exploration and refinement.

Further Scope

Solar Irradiance:

Solar irradiance refers to the power per unit area received from the Sun in the form of electromagnetic radiation. It's typically measured in watts per square meter (W/m^2). This value varies depending on factors such as time of day, time of year, weather conditions, and geographic location. This can be included by representing a Vector I , representing intensity of light, in the same direction as Vector S_{sun} . The magnitude of this vector, $|I|$, represents the magnitude of light intensity in W/m^2 .

Axial tilt:

As stated earlier, the axial tilt of earth is actually 23.5° . This causes the existence of different seasons across a year and different sunrise and sunset times. Hence, if we were to find the optimal tilt angle for one year, we would have to take into account the varying factors. It is unfeasible to change the tilt angle every day, hence the methodology of finding optimum tilt angle for a certain day would no longer work.

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